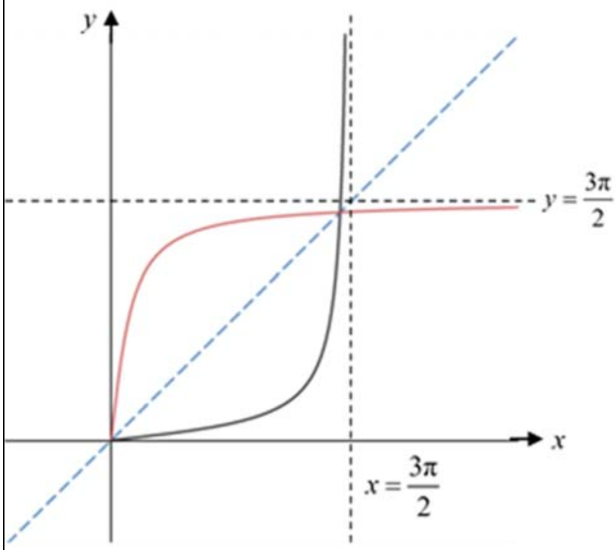
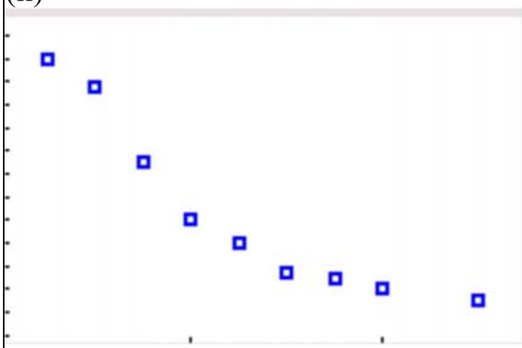
ANNEX B

CJC H2 Math JC2 Preliminary Examination Paper 2

QN	Topic Set	Answers
1	Equations and Inequalities	$y=\frac{1}{3}x^3-\frac{1}{2}x^2-6x+4$
2	Vectors	(ii) $\sqrt{2}$ (iv) $\frac{5\sqrt{2}}{2}$ unit^2
3	Functions	(i)  (ii) $y=x$ (iii) $\alpha^2+2\ln\left[\cos\left(\frac{\alpha}{3}\right)\right]$ (iv) $\mathrm{gf}:x\mapsto \mathrm{e}^{\frac{1}{3}\tan\left(\frac{x}{3}\right)}$ for $x\in\mathbb{R}$, $0\le x<\frac{3\pi}{2}$
4	Complex numbers	(a) $w=3+4\mathrm{i}$, $z=7-\mathrm{i}$ $w=-4+4\mathrm{i}$, $z=14-\mathrm{i}$ (b)(i) $3-\mathrm{i}$ (ii) $w=1-2\mathrm{i}$, $w=-1-3\mathrm{i}$ (c)(i) $\frac{1}{2a^3}\mathrm{e}^{\mathrm{i}\left(-\frac{3\pi}{4}\right)}$ (ii) 2, 6
5	P&C, Probability	(i) 100 (ii)7680 (iii)92160

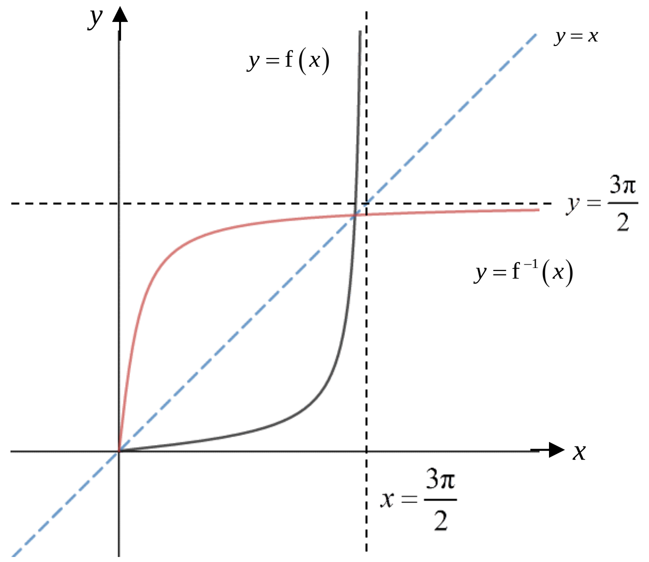
6	Correlation & Linear Regression	(ii)  (iv) $a = 0.257$, $b = 22.8$ (v) 1860kg (vi)(a) $\overline{y}$ will be increased by a. (b) remain unchanged. (c) remain unchanged.
7	Binomial Distribution	(a)(i) $p = \dfrac{5}{11}$ (ii) $p = \dfrac{1}{2}$ (b)(i) $= 0.0000514$ (ii) 0.998
8	Normal Distribution	(i) 1.31%. (ii) $t = 307.51\text{mm}$ (iii)0.525
9	Hypothesis Testing	(ii) $\overline{t} = 47.69$ thousand hours $s^2 = 44.3$ (iv)Yes
10	DRV	(ii) $x = 5$ (iv) $\dfrac{127}{252}$ or 0.504 (v) $\dfrac{11}{127}$

Q1. System of Linear Equations		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Formulate a system of linear equations. Solve a system of linear equations using a G.C.	**(i)** $y = ax^3 + bx^2 + cx + d$ Curve passes through $\left(-2, \frac{34}{3}\right)$: $a(-2)^3 + b(-2)^2 + c(-2) + d = \frac{34}{3}$ $-8a + 4b - 2c + d = \frac{34}{3}$ — ① Curve passes through $\left(3, -\frac{19}{2}\right)$: $a(3)^3 + b(3)^2 + c(3) + d = -\frac{19}{2}$ $27a + 9b + 3c + d = -\frac{19}{2}$ — ② $\frac{\mathrm{d}y}{\mathrm{d}x} = 3ax^2 + 2bx + c$ Curve has maximum point $\left(-2, \frac{34}{3}\right)$: $3a(-2)^2 + 2b(-2) + c = 0$ $12a - 4b + c = 0$ — ③ Curve has minimum point $\left(3, -\frac{19}{2}\right)$: $3a(3)^2 + 2b(3) + c = 0$ $27a + 6b + c = 0$ — ④	Most common mistake: - Some students assumed the coeff of x^3 is 1, eg, $y = x^3 + bx^2 + cx + d$ Some attempt to form ONLY 2 or 3 equations to solve for 4 unknowns; note that at least 4 eqns are needed to solve for 4 unknowns. A few students left their eqn as $\frac{1}{3}x^3 - \frac{1}{2}x^2 - 6x + 4$ instead of $y = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 6x + 4$

	Solving, $a=\frac{1}{3}, b=-\frac{1}{2}, c=-6, d=4$ $\therefore\ y=\frac{1}{3}x^3-\frac{1}{2}x^2-6x+4$	

Q2. Vectors		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Concept of geometrical interpretation.	**(i)** Length of projection of $\mathbf{a}$ on to $\mathbf{b}$	Generally OK, but many gave the answer as length of projection of $\mathbf{b}$ onto $\mathbf{a}$.
Concept of unit vector and cross product formula.	**(ii)** $\lvert\mathbf{a}\times\mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$ $= (2)(1)\sin\frac{\pi}{4}$ $= \sqrt{2}$	Many students mixed up the definition of dot and cross product, although $\sin\frac{\pi}{4}$ is the same as $\cos\frac{\pi}{4}$ which some students ended up with the correct final answer, but they still get penalized as they are using the wrong definition.
Expansion of cross product.	**(iii)** $\mathbf{p}\times\mathbf{q}$ $= \left[3\mathbf{a}+(\mu+2)\mathbf{b}\right]\times\left[(\mu+3)\mathbf{a}+\mu\mathbf{b}\right]$ $= 3(\mu+3)(\mathbf{a}\times\mathbf{a})+3\mu(\mathbf{a}\times\mathbf{b})+\left(\mu^2+5\mu+6\right)(\mathbf{b}\times\mathbf{a})+\mu(\mu+2)(\mathbf{b}\times\mathbf{b})$ $= \left(-3\mu+\mu^2+5\mu+6\right)(\mathbf{b}\times\mathbf{a})$ $[\because \mathbf{a}\times\mathbf{a}=\mathbf{0}$ **and** $\mathbf{b}\times\mathbf{b}=\mathbf{0}\ ]$ $= \left(\mu^2+2\mu+6\right)(\mathbf{b}\times\mathbf{a})$	Common mistakes: - $\mathbf{a}\times\mathbf{a}=\lvert\mathbf{a}\rvert^2$ - The third term in the expansion was $\left(\mu^2+5\mu+6\right)(\mathbf{a}\times\mathbf{b})$ instead of $\left(\mu^2+5\mu+6\right)(\mathbf{b}\times\mathbf{a})$; note that the direction of cross product is important. - $\underset{\sim}{a}\times\underset{\sim}{a} = \underset{\sim}{0} \rightarrow$ null vector not $\underset{\sim}{a}\times\underset{\sim}{a} = 0$
Finding stationary value.	**(iv)** Area $OPQ = \frac{1}{2}\left\lvert\left(\mu^2+2\mu+6\right)\right\rvert\left\lvert(\mathbf{b}\times\mathbf{a})\right\rvert$ $=\frac{1}{2}\left\lvert\left(\mu^2+2\mu+6\right)\right\rvert\sqrt{2}$ $=\frac{\sqrt{2}}{2}\left\lvert(\mu+1)^2+5\right\rvert$	Common mistake: $\frac{1}{2}\left(\mu^2+2\mu+6\right)\mathbf{b}\times\mathbf{a}$ Note that the above expression is a vector, not magnitude.

	Smallest Area $OPQ = \ = \dfrac{5\sqrt{2}}{2}\,\text{unit}^2$	

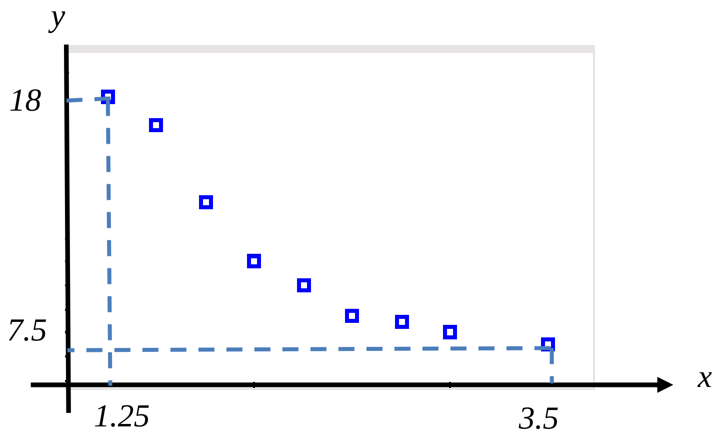
<table>
<tr><th colspan="3">Q3. Functions & Definite Integrals</th></tr>
<tr><th>Assessment Objectives</th><th>Solution</th><th>Examiner’s Feedback</th></tr>
<tr><td rowspan="2">Understand the relationship between a function and its inverse.</td><td>(i)

</td><td>Many students did not fully extend the curve past $x=\frac{3\pi}{2}$ and/or $y=\frac{3\pi}{2}$</td></tr>
<tr><td>(ii) $y=x$</td><td></td></tr>
<tr><td>Find area bounded by 2 curves.</td><td>(iii) <u>Method ①:</u>
$\text{Area } = 2\int_0^{\alpha} x-\frac{1}{3}\tan\left(\frac{x}{3}\right)\mathrm{d}x$
$=2\left[\frac{x^2}{2}-\ln\left|\sec\left(\frac{x}{3}\right)\right|\right]_0^{\alpha}$
$=2\left[\frac{\alpha^2}{2}-\ln\left|\sec\left(\frac{\alpha}{3}\right)\right|-0+0\right]$
$=\alpha^2+2\ln\left[\cos\left(\frac{\alpha}{3}\right)\right]$</td><td>Most students did not use this method, opting for the more tedious alternative.
Students can use area of triangle formula $\frac{1}{2}\alpha(\alpha)$ instead of $\int_0^{\alpha} x\,\mathrm{d}x$
(Some used $\frac{1}{2}\alpha\left(\frac{1}{3}\tan\left(\frac{\alpha}{3}\right)\right)$
which is not simplified</td></tr>
</table>

	Method ②: $$\text{Area } = \int_0^{\alpha} 3\tan^{-1}3x - \frac{1}{3}\tan\left(\frac{x}{3}\right)\mathrm{d}x$$ $$= \left[3x\tan^{-1}3x\right]_0^{\alpha} - \int_0^{\alpha} 3x\frac{3}{1+(3x)^2}\,\mathrm{d}x - \left[\ln\left(\sec\frac{x}{3}\right)\right]_0^{\alpha}$$ $$= 3\alpha\tan^{-1}3\alpha - \frac{1}{2}\int_0^{\alpha}\frac{18x}{1+9x^2}\,\mathrm{d}x - \ln\left(\sec\frac{\alpha}{3}\right) + \ln 1$$ $$= 3\alpha\tan^{-1}3\alpha - \frac{1}{2}\left[\ln\left(1+9x^2\right)\right]_0^{\alpha} - \ln\left(\sec\frac{\alpha}{3}\right)$$ $$= 3\alpha\tan^{-1}3\alpha - \frac{1}{2}\ln\left(1+9x^2\right) - \ln\left(\sec\frac{\alpha}{3}\right)$$	Note the two answers are equal. Most students did this method, not utilizing the symmetry of the curves.
Determine if the composite function exists. Find the rule and domain of a composite function.	**(iv)** $\mathrm{R_f} = [0, \infty)$ $\mathrm{D_g} = [-2, \infty)$ Since $\mathrm{R_f} \subseteq \mathrm{D_g}$, gf exists. $$\mathrm{gf}(x) = \mathrm{g}\left[\frac{1}{3}\tan\left(\frac{x}{3}\right)\right]$$ $$= \mathrm{e}^{\frac{1}{3}\tan\left(\frac{x}{3}\right)}$$ $$\mathrm{D_{gf}} = \mathrm{D_f} = \left[0, \frac{3\pi}{2}\right)$$ $\mathrm{gf} : x \mapsto \mathrm{e}^{\frac{1}{3}\tan\left(\frac{x}{3}\right)}$ for $x \in \mathbb{R}$, $0 \le x < \dfrac{3\pi}{2}$.	Common mistakes: $\mathrm{D_{gf}} = \mathrm{D_g} = [-2, \infty)$ Many students did not put in similar form

Q4. Complex Numbers		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Solving simultaneous equations involving complex numbers.	**(a)** $z = 10 - w^* - 5\text{i}$ $\lvert w\rvert^2 = 10 - w^* - 5\text{i} + 18 + \text{i}$ $\lvert w\rvert^2 + w^* = 28 - 4\text{i}$ Let $w = a + b\text{i}$, $a^2 + b^2 + a - b\text{i} = 28 - 4\text{i}$ By comparing, $b = 4$, $a^2 + (4)^2 + a = 28$ $a^2 + a - 12 = 0$ $(a+4)(a-3) = 0$ $a = -4$ or $a = 3$ $\therefore w = 3 + 4\text{i}$ or $w = -4 + 4\text{i}$ When $w = 3 + 4\text{i}$, $z = 10 - (3 - 4\text{i}) - 5\text{i}$ $= 7 - \text{i}$ When $w = -4 + 4\text{i}$, $z = 10 - (-4 + 4\text{i}) - 5\text{i}$ $= 14 - 9\text{i}$	Most students were able to do this question except for the occasional slips in algebraic manipulation. A number of students mistook $\lvert w\rvert^2$ for w^2. Presentation for simultaneous equation is unclear.
	(b)(i) $z^2 - 5z + 7 + \text{i} = [z - (2+\text{i})][z - k]$ By comparing coefficient of z: $z^2 - 5z + 7 + \text{i} = [z - (2+\text{i})][z - k]$ $-5 = -k - (2 + \text{i})$ $k = 3 - \text{i}$ The second root is $3 - \text{i}$	Generally well done.

	(ii) $-\mathrm{i}w^2+5w+7\mathrm{i}-1=0$ $-w^2-5\mathrm{i}w+7+\mathrm{i}=0$ $(\mathrm{i}w)^2-5(\mathrm{i}w)+7+\mathrm{i}=0$ $\mathrm{i}w=2+\mathrm{i}$ or $\mathrm{i}w=3-\mathrm{i}$ $w=1-2\mathrm{i}$ or $w=-1-3\mathrm{i}$	Badly done. Most students fail to identify the term to replace.
Property of modulus and argument.	**(c)(i)** **<u>Method ①:</u>** $z=-a+a\mathrm{i}$ $=a\sqrt{2}\mathrm{e}^{\mathrm{i}\left(\frac{3\pi}{4}\right)}$ $w=\left(\mathrm{e}^{\mathrm{i}(\pi)}\right)\dfrac{\sqrt{2}a\sqrt{2}\mathrm{e}^{\mathrm{i}\left(-\frac{3\pi}{4}\right)}}{4a^4\mathrm{e}^{\mathrm{i}(3\pi)}}$ $=\dfrac{1}{2a^3}\mathrm{e}^{\mathrm{i}\left(-\frac{11\pi}{4}+2\pi\right)}$ $=\dfrac{1}{2a^3}\mathrm{e}^{\mathrm{i}\left(-\frac{3\pi}{4}\right)}$	Badly done. Most students prefer to simplify the denominator but had problems with the algebraic manipulation. Most students got the argument wrong as they left their answer as $-\dfrac{1}{2a^3}\mathrm{e}^{\mathrm{i}\alpha}$, or they mistook $\arg(-\sqrt{2}z^*)=-\sqrt{2}\arg(z^*)$. Another common mistake was that many students left the argument of z as $\frac{\pi}{4}$.
Concepts of argument to find purely imaginary roots.	**(ii)** If $\mathrm{Re}\,(w^n)=0$, $n\left(-\dfrac{3\pi}{4}\right)=\dfrac{\pi}{2}+k\pi,$ where $k\in\mathbb{Z}$ $n=-\dfrac{2}{3}(1+2k),$ where $k\in\mathbb{Z}$ Three smallest positive whole number values of n are 2, 6.	Most students got the method marks but lost the last mark as their argument was wrong.

Q5. Permutations and Combinations		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Solving simple counting problems involving multiplication principle	**(i)** No. of ways $= {}^{10}C_1 \times {}^{10}C_1 = 100$	Generally well done except for some who did ${}^{10}C_1 + {}^{10}C_1$ instead.
Solving counting problems involving circular arrangements	**(ii)** No. of ways $= (6-1)! \times (2!)^6 = 7680$	Some students used $(6-1)! \times 2!$ or $(6-1)! \times 6(2)!$ instead. Since there are 6 couples, and for each couple there are 2! ways to arrange them, we do $2! \times 2! \times 2! \times 2! \times 2! \times 2!$, which is different from $6(2)!$.
Solving complex arrangement problems involving restrictions	**(iii)** **Method ①:** (1) Arrange P and GoH $= 2!$ (2) Choose side where CP and Sec are on (left or right) $= {}^2C_1$ (3) Choose 2 classes to be seated with CP and Sec $= {}^5C_2$ (4) Arrange the 2 classes and the people within each class $= 2! \times (2!)^2$ (5) Slot in CP and Sec $= {}^3C_2 \times 2!$ (6) Arrange the 3 other classes and the people within each class $= 3! \times (2!)^3$ No. of ways $= 2! \times 2 \times {}^5C_2 \times 2! \times (2!)^2 \times {}^3C_2 \times 2! \times 3! \times (2!)^3 = 92160$ **Method ②: (Arrange the 5 classes at one go)** No. of ways $= 2! \times 2 \times 5! \times (2!)^5 \times {}^3C_2 \times 2! = 92160$ **Method ③: (Complement)** No. of ways $=$ n(CP/S on 1 side but may be tog) $-$ n (CP/S tog) $= \left\{2 \times \left[{}^5C_2 \times 3! \times 2^3\right] \times 2 \times (2)^2 \times 4!\right\} - \left[6! \times (2)^6 \times 2\right] = 92160$	Most students were able to get 1 or 2 marks for this part. Key is for the GoH and P to be seated in the middle, and for the CP and S to be separated, they must be on the same side. Otherwise with 5 students on one side and 7 students on the other side, GoH and P will not be in the middle. So the remaining 5 classes will be split to 3-2, with CP and S joining the side with 2 classes. Hence, using the 2 classes, we will choose 2 out of 3 slots for CP and S. Bear in mind that the 2 classes can either be on the left, or on the right.

Q6. Correlation and Linear Regression		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Concept of $(\bar{x}, \bar{y})$ lies on the regression line.	**(i)** $\bar{y} = 22.51355 - 4.908387\bar{x}$ $\left(\frac{91+k}{9}\right) = 22.51355 - 4.908387\left(\frac{20.5}{9}\right)$ $k = 11.0$	Poorly attempted. Many students simply substituted $x = 2$ into the equation of the regression line and hoped that the resulting y, i.e. k will be 2. They failed to understand that the point $x = 2$ may not pass through the regression line. Students must understand the concept that $(\bar{x}, \bar{y})$ lies on the regression line.
Scatter diagram	**(ii)** 	Most students handled this part accurately. There were students who carelessly wrote the x-intercepts as y-intercepts and vice versa. Others thought that 1.25 > 3.5 and 7.5 > 18. All these could have been avoided if students made an effort to check their scatter diagram before proceeding.

Concept of linearization of non-linear model.	**(iii)**	As x increases, y decreases at a decreasing rate and tends towards a limit.	Poorly attempted. Students merely says that the graph of model (C) is similar to the graph in the scatter diagram. This warrants no marks. Students are reminded that they need to describe the shape of the graph. Students are advised against describing the gradient of the scatter diagram as it is prone to careless mistakes. In this question, as x increases, the gradient actually increases because it becomes less negative.
Use of GC to find the regression line.	**(iv)**	$a = 0.257$ $b = 22.8$	Many students failed to leave their final answer in 3 s.f.
Estimation and its reliability.	**(v)**	$y = 0.25681 + \dfrac{22.837}{x}$ $12 = 0.25681 + \dfrac{22.837}{x}$ $x = 1.94$ (3 s.f.) The weight of the car is 1940kg. The prediction is reliable as $y = 12$ is within the data range of y and the $\|r\|$-value is close to 1.	Many students left their answer as x = 1.94. They did not conclude that the weight of the car is 1940kg or 1.94 kg in thousands. Many students failed to mention that the $\|r\|$-value is close to 1 when stating that the prediction is reliable.
Concept of mean and standard deviation	**(vi)**	(a) $\bar{y}$ will be increased by a. (b) Standard deviation of y remain unchanged. (c) Correlation coefficient remain unchanged.	Well-attempted by students.

Q7. Binomial Distribution and Sampling Distribution		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Setting up and solving equations using the formula for a Binomial random variable	**(a)(i)** $\mathrm{P}(X=4)=\mathrm{P}(X=5)$ $\frac{10!}{4!6!}p^4(1-p)^6=\frac{10!}{5!5!}p^5(1-p)^5$ $5(1-p)=6p$ $p=\frac{5}{11}$	Most students could identify that the probabilities for the outcomes of 4 and 5 should be equal and wrote the expressions according to the formula. Some failed to solve the equation due to inadequate skills in algebra.
Setting up and solving equations using the formula for a Binomial random variable	**(ii)** $\mathrm{P}(X\le 9)=\frac{1023}{1024}$ $\mathrm{P}(X=10)=\frac{1}{1024}$ $p^{10}=\left(\frac{1}{2}\right)^{10}$ $p=\frac{1}{2}$	Many students did not realise that the complementary case is simply 10. Once this hurdle was overcome most were able to find the final answer.
Solving simple problems based on random samples from a binomial random variable	**(b)(i)** $\mathrm{P}(Y\le 256)=0.719485301$ P(all 100 values are less than or equal to 256) $=0.719485301^{30}$ $=0.0000514$	Many students immediately dived into the irrelevant routine of using CLT to find the sampling distribution once they saw the conditions given, without analyzing the question carefully. Majority of the students left the first probability as the answer. Their understanding of the term "sample" may be in question.

Applying Central Limit Theorem for the sampling distribution of a random sample from a binomial random variable	**(ii)** $\mathrm{E}(Y)=500(0.5)=250$, and $\mathrm{Var}(Y)=500(0.5)(0.5)=125$ Since the sample size is sufficiently large, $\bar{Y} \sim \mathrm{N}\left(250, \dfrac{125}{30}\right)$ approximately by CLT $\mathrm{P}\left(\bar{Y} \le 256\right)=0.998$	Most were able to follow the routine to write down the expectation and variance of Y. However, half of them did not show clear understanding of sampling distribution and central limit theorem in their subsequent presentation of the solution. The most common mistake is that quoting CLT to write down $Y \sim N(250,125)$, which is WRONG! It is the mean of samples of large size may be considered as normally distributed approximately, not the individual observation. Other common mistakes include forgetting to divide the variance by sample size, or using a wrong notation for the random variable of sample mean.
Making comparison between probabilities that are calculated based on the context	**(iii)** The probability in part **(ii)** included cases where some of the values can be larger than 256, but the final average is still at most 256.	Many students were able to give the correct reason though the phrasing can still be improved. For example many casually wrote "probability in (i) is a subset of probability in (ii)", which showed understanding but failed to make sense mathematically when it is the collection of "events/outcomes" in one being subset of the other.

Q8. Normal Distribution		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Practical application of Normal distribution to obtain percentages of items with certain properties	**(i)** Let L be the random variable denoting length of a trading card in mm. $L \sim \mathrm{N}(89,\ 0.2025)$ $\mathrm{P}(L > 90) = 0.0131$, hence the percentage is 1.31%.	Badly done by students. Mistakes: 1) Confusing CRV and DRV by writing $\mathrm{P}(L \le 90) = \mathrm{P}(L \le 89)$ 2) Not answering in %
Practical application of Normal distribution to obtain the critical value of a certain property satisfied by a given percentage of items	**(ii)** Let T be the random variable denoting the perimeter of a trading card, in mm. $T \sim \mathrm{N}\left(2(64)+2(89),\ 2^2(0.3^2)+2^2(0.45^2)\right)$ $\sim \mathrm{N}\left(306,\ 1.17\right)$ $\mathrm{P}(T > t) = 0.08$ $\mathrm{P}(T < t) = 0.92$ Hence $t = 307.51\mathrm{mm}$	Badly done by students. Most common mistake: Taking invNorm with RHS area 0.2.
Practical application of Normal distribution to obtain probability of randomly selected items fulfilling certain independent physical properties	**(iii)** Let X and Y be random variable denoting the width and length of a card sleeve subtracting away the width and length of a trading card respectively in mm. Hence $X \sim \mathrm{N}\left(66-64,\ (0.45^2)+(0.3^2)\right)$ $\sim \mathrm{N}\left(2,\ 0.2925\right)$ and $Y \sim \mathrm{N}\left(91-89,\ (0.45^2)+(0.675^2)\right)$ $\sim \mathrm{N}\left(2,\ 0.658125\right)$ $\mathrm{P}(\text{width fits nicely}) = \mathrm{P}(0 < X \le 2.4) = 0.7701200999$ $\mathrm{P}(\text{length fits nicely}) = \mathrm{P}(0 < Y \le 2.4) = 0.6821730404$ $\mathrm{P}(\text{sleeve fits nicely}) = \mathrm{P}(\text{width fits nicely})\,\mathrm{P}(\text{length fits nicely})$ $= 0.7701200999 \times 0.6821730404$ $= 0.525$	Badly done. Better students were able to find the new mean and new variance but mistakes were made when calculate the probabilities, Mistakes: 1) $\mathrm{P}(X \le 2.4)\,\mathrm{P}(Y \le 2.4)$ 2) $\mathrm{P}(X \le 1.2)\,\mathrm{P}(Y \le 1.2)$ 3) $\mathrm{P}(0 \le X \le 1.2)\,\mathrm{P}(0 \le Y \le 1.2)$ Most students left this part blank.

Q9. Hypothesis Testing		
Assessment Objectives	**Solution**	**Examiner's Feedback**
Provide reasoning to support the application of Central Limit Theorem in Hypothesis testing	**(i)** It is not necessary as the sample size is sufficiently large for Central Limit Theorem to apply.	This part is poorly attempted with many students discussing about the PARAMETERS of the distribution rather than the fact of whether it is a normal distribution. Some students simply stated that it is necessary as the hypothesis testing requires the use of a normal distribution, showing clearly their lack of understanding for the Central Limit Theorem and Sampling Distributions in general. There are also quite a number of students who either left out the fact that the sample size is large or that Central Limit Theorem is applicable, resulting in an incomplete explanation.
Conduct a z-test for a practical situation	**(ii)** $\bar{t} = 2384.5/50 = 47.69$ thousand hours $s^2 = \frac{1}{50-1}\left(115885.23 - \frac{2384.5^2}{50}\right) = 44.25357143 = 44.3$ $\mathrm{H}_0 : \mu = 50$ $\mathrm{H}_1 : \mu \neq 50$ Under H_0, since n is large, by C.L.T. $\bar{T} \sim \mathrm{N}\left(50, \frac{44.25357143}{50}\right)$ appx p-value = 0.01407 OR test statistic = – 2.4554	Most candidates are successful with the unbiased estimates, but some left the answer as 47.7 for population mean, not realizing that it is an exact decimal. Quite a number of students also quoted the wrong formula for population variance or left their answers as the value before dividing by 49. While most students with the correct estimates were successful

		Since p-value = 0.01407 > 0.01, we do not reject H_0 and conclude that there is insufficient evidence at 1% level of significance to claim that the mean number of hours before failure is not 50 thousand hour.	with the testing, there is also a significant number of students who lost all marks by simply stating an incorrect p-value based on their wrong parameters. P-values calculated based on 3 s.f. values of the parameters or a combination of 5 s.f. and 3 s.f. values were accepted. Correct p-values based on erroneous presentation of the sampling distributions were not penalized due to benefit of doubt given. Students who attempted to use critical values were less successful as they applied modulus to the the test statistic without doing so for the critical value resulting in erroneous comparisons. Students should state clearly the rejection region when using critical values. Most conclusions were not given in context, did not mention level of significance, or were not phrase in terms of the alternate hypothesis. Many students phrased the conclusion wrongly as "having sufficient evidence to claim that the mean is 50 thousand hours".
Provide reasoning to support the choice of alternate hypothesis based on the context of the practical situation	**(iii)**	The reader would be more interested to test whether the mean is actually lower than the stated value which is not beneficial to them.	Most students simply stated that the test did not indicate whether the mean is more of less, but did not make any reference to why these

		cases would matter to the reader. Students who manage to make reference to higher mean being beneficial and/or lower mean being not beneficial, were given credit based on benefit of doubt.
Identifying alternate hypothesis relevant to the context and provide reasoning on the effect of the alternate hypothesis on the resulting p-value and final conclusion of a z-test	**(iv)** $H_1 : \mu < 50$ Yes, the result will differ as the p-value will be halved when switching to a one-tail test.	Most students were successful with stating the correct alternate hypothesis, except some who used the left tail test in (ii). However, not all were able to provide an explanation to support the change in conclusion, especially those who used critical values in (ii). Students who did not identify that the p-value is exactly half of the value found in (ii) will have to state the actual value, simply mentioning that the p-value is smaller is in sufficient as 0.011 is also a smaller p-value, but it will not result in a change in the conclusion. Most students who re-did the test were most successful for this part, but many went on to write the full conclusion, which is not required by the question.
Identifying implicit assumptions made in a hypothesis test	**(v)** We need to assume that the usage hours before failure for hard drives are independent for all hard drives.	Many students were able to state the assumption needed, but some did not exhibit any understanding of the situation.

<table>
<tr><th colspan="3">Q10. Probability and Discrete Random Variables</th></tr>
<tr><th>Assessment Objectives</th><th>Solution</th><th>Examiner's Feedback</th></tr>
<tr><td>Formulating an expression for the probability density function of a discrete random variable</td><td>

(i)

$$\begin{aligned}&P(S=6)\\&=P(RRBBB)+P(RGGGB)\\&=\frac{{}^5C_3(2!)\left[x(x-1)(x-2)\right]}{(x+5)(x+4)(x+3)(x+2)(x+1)}+\frac{{}^5C_3x(2)(2)(3!)}{(x+5)(x+4)(x+3)(x+2)(x+1)}\\&=\frac{20x\left[x^2-3x+2\right]}{(x+5)(x+4)(x+3)(x+2)(x+1)}+\frac{20x(12)}{(x+5)(x+4)(x+3)(x+2)(x+1)}\\&=\frac{20x(x^2-3x+14)}{(x+5)(x+4)(x+3)(x+2)(x+1)}\end{aligned}$$

</td><td>This part is usually well done.
Lack of essential working is not acceptable.</td></tr>
<tr><td>Solving for an unknown parameter based on the expression for the probability density function of the discrete random variable</td><td>

(ii)

$$\frac{5}{63}=\frac{20x(x^2-3x+14)}{(x+5)(x+4)(x+3)(x+2)(x+1)}$$

$$5(x+5)(x+4)(x+3)(x+2)(x+1)-1260x(x^2-3x+14)=0$$

Solving, the only integer root is $x=5$

</td><td>This part is well done.</td></tr>
<tr><td>Completing the probability distribution table of a discrete random variable</td><td>

(iii)

s	1	4	7	25
$P(S=s)$	$\frac{5}{84}$ or $\frac{15}{252}$	$\frac{5}{21}$ or $\frac{60}{252}$	$\frac{5}{42}$ or $\frac{30}{252}$	$\frac{1}{252}$

$P(S=1)=P(\text{GBBBB})$

$$=\left(\frac{3}{10}\right)\left(\frac{5}{9}\right)\left(\frac{4}{8}\right)\left(\frac{3}{7}\right)\left(\frac{2}{6}\right)\left(\frac{5!}{4!}\right)$$

$$=\frac{5}{84}\text{ or }\frac{15}{252}$$

Note: $\frac{5!}{4!}$ is for arranging GBBBB with 4 repeated "B"s.

</td><td>$P(S=4)$ and $P(S=7)$ proved to be quite challenging for most candidates.</td></tr>
</table>

	$\mathrm{P}(S=4)=\mathrm{P}(\mathrm{RGBBB})$ $=\left(\frac{2}{10}\right)\left(\frac{3}{9}\right)\left(\frac{5}{8}\right)\left(\frac{4}{7}\right)\left(\frac{3}{6}\right)\left(\frac{5!}{3!}\right)$ $=\frac{5}{21}$ or $\frac{60}{252}$ Note: $\frac{5!}{3!}$ is for arranging RGBBB with 3 repeated "B"s. $\mathrm{P}(S=7)=\mathrm{P}(\mathrm{RRGBB})$ $=\left(\frac{2}{10}\right)\left(\frac{1}{9}\right)\left(\frac{3}{8}\right)\left(\frac{5}{7}\right)\left(\frac{4}{6}\right)\left(\frac{5!}{2!2!}\right)$ $=\frac{5}{42}$ or $\frac{30}{252}$ Note: $\frac{5!}{2!2!}$ is for arranging RRGBB with 2 repeated "R"s and 2 repeated "B"s. $\mathrm{P}(S=25)=\mathrm{P}(\mathrm{BBBBB})$ $=\left(\frac{5}{10}\right)\left(\frac{4}{9}\right)\left(\frac{3}{8}\right)\left(\frac{2}{7}\right)\left(\frac{1}{6}\right)$ $=\frac{1}{252}$	
Calculating expectation of a discrete random variable and probabilities based on the probability distribution table	**(iv)** $\mathrm{E}(S)=4.60$ or $\frac{1159}{252}$ $\mathrm{P}(S>4.60)=\frac{127}{252}$ or 0.504	This part is poorly done as a result of errors from (iii).

Solving for conditional probabilities based on a discrete random variable	**(v)**	$\mathrm{P}(\text{no}\,G \mid S > 4.60)$ $= \dfrac{\mathrm{P}(RRBBB) + \mathrm{P}(BBBBB)}{\frac{127}{252}}$ $= \dfrac{\frac{20(5)(4)(3)}{(10)(9)(8)(7)(6)} + \frac{1}{252}}{\frac{127}{252}}$ $= \dfrac{\frac{10}{252} + \frac{1}{252}}{\frac{127}{252}}$ $= \dfrac{11}{127}$	Common omission or error pertaining to the case $\mathrm{P}(RRBBB)$, led to wrong answer. This part is poorly done.